No. of Functional Features Included in the Solution

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Interactive Q&A Service	Answers complex academic questions with simplified explanations using Gemini API.	/qa & /explain Endpoints	Done	1.0 Mark
2	Automated Quiz Generator	Generates custom multiple-choice practice quizzes based on topics or uploaded text.	/quiz Endpoint	Done	1.0 Mark
3	Text & Document Summarizer	Condenses long educational articles and textbook passages into concise bullet points.	/summarize Endpoint	Done	1.0 Mark
4	Personalized Learning Roadmaps	Provides structured, step-by-step learning paths for specified topics or skills.	/learn/recommEndations Endpoint	Done	0.5 Mark
5	Offline Local Model Fallback	Performs basic offline Q&A using LaMini-Flan-T5-783M when internet/cloud API is unavailable.	Core AI Logic Service	Done	0.5 Mark
6	Interactive Web Dashboard	User-friendly UI with Jinja2 templates for inputting prompts and viewing formatted outputs.	Presentation / Client Layer	Done	0.5 Mark
7	RESTful API Architecture	Asynchronous route processing for handling multiple concurrent user	FastAPI / Uvicorn Server	Done	0.25 Mark

		requests efficiently.			
8	Environment Configuration & Key Management	Securely isolates Gemini API credentials and runtime settings using .env handling.	Backend Security Layer	Done	0.25 Mark

Feature Summary:

Metric	**Count / Value**
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	**Category**	**Features in Category**	**Example Features**
1	User Interface (UI)	1	Jinja2 Web Dashboard for interactive inputs and dynamic result displays.
2	Backend / Logic	3	FastAPI Async Routers, Dynamic Prompt Formatting, Offline LaMini-Flan-T5 Fallback Pipeline.
3	Database / Storage	1	Stateless In-Memory Session Cache & Local File Directory Storage.
4	API / Integration	2	Google Gemini 1.5 Pro API SDK Integration and Modular REST API Endpoints (/qa, /quiz, /summarize).
5	Security / Authentication	1	API Key Protection via Environment Variables and Stateless Request Validation.